DA4022 — Data Analytics & Applications | Mid-Project Report | October 2026

UCA: Uncertainty-Calibrated Adaptive Compression for LLM Inference

Core Research Question:
Can we design a better adaptive compression system, inspired by C-DIC and HyC-LoRA, that uses uncertainty signals (Internal Confidence + SemanticDPP) to dynamically control how much and what kind of compression to apply at inference time — and beat existing benchmarks?

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DA4022 Course Project · Topic Anchor #8
"Can we use uncertainty for optimal memory allocation?"

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1. Executive Summary

Current Large Language Model (LLM) serving systems allocate a *fixed* computational and memory budget — the same maximum KV cache size, full context window, and static decoding strategy — regardless of query complexity. This leads to two pathologies: (1) **resource waste** on simple, high-confidence queries, and (2) **hallucination risk** on ambiguous, knowledge-intensive prompts where insufficient resources cause the model to confabulate.

This project proposes **UCA (Uncertainty-Calibrated Adaptive Compression)**, a novel inference middleware that synthesises four recent ICLR/ICML/MLSys 2025–2026 papers to convert uncertainty signals into *continuous, per-dimension compression ratios* rather than the hard binary switches used by all existing systems.

> 🎯 **The Core Insight**
>
> **No existing method uses uncertainty as a continuous control signal for compression.** All four papers either estimate uncertainty without acting on it (IC, SemanticDPP), or compress without consulting uncertainty (C-DIC, HyC-LoRA). UCA bridges this gap by making the compression ratio itself a differentiable function of uncertainty.

UCA uses a fast IC pre-filter so that the expensive DPP computation runs only on the truly uncertain subset (~40% of queries), achieving a ~3× speedup over full SemanticDPP while targeting AUROC >0.83 with >32% KV cache reduction — beating all four baseline methods simultaneously.

2. Background: Critical Analysis of Four Reference Papers

2.1 Paper A — Query-Level Uncertainty in LLMs (Chen et al., ICLR 2026)

Code: `tigerchen52/query_level_uncertainty` | *Venue: ICLR 2026*

What It Does

Internal Confidence (IC) is a training-free uncertainty estimator requiring only a single forward pass over the input query — no tokens are generated. It probes P(YES) logits at every (layer l, token position t) pair and aggregates via **Attenuated Encoding**:

$$IC(q) = \Sigma_{l,t}\ w(l,t) \cdot P(\text{YES} \mid h_l^{(t)}),\ w(l,t) \propto \exp(-\lambda \cdot \text{dist}(l,t)) \qquad (1)$$

where *dist(l,t)* is the distance from the "optimal decision centre" (last token, last layer). Achieves **30–600× speedup** over answer-level methods.

Critical Gaps We Target

- **Binary routing only:** IC uses a hard threshold τ — binary decision only, no continuous resource modulation.
- **Static fallback:** Low IC always triggers RAG or cascade; never controls *how aggressively* to compress context.
- **Threshold is dataset-specific:** Optimal τ and k (last-k tokens) must be re-tuned per dataset/model pair.

2.2 Paper B — SemanticDPP (Hamer et al., ICLR 2026)

Authors: Jenny Hamer, Jasper Snoek, et al. (Google DeepMind)

What It Does

Samples *m* responses $S = \{r_1,\ldots,r_m\}$ and measures semantic diversity via a Determinantal Point Process (DPP) L-ensemble over projected output embeddings. Learns projection ϕ: $\mathbb{R}^d \to \mathbb{R}^p$ ($p \ll d$). The kernel and uncertainty score are:

$$K(x_i, x_j) = \exp(-\alpha\|\phi(x_i)-\phi(x_j)\|^2),\quad U(S) = \log\det(L_S) - \log\det(L+I) \tag{2}$$

Critical Gaps We Target

- **Requires m ≥ 5 generations:** Expensive for real-time routing.
- **Discards kernel structure:** U(S) is a scalar; all rich pairwise information in L_S is lost after log-det.
- **No compression action:** Pure estimation; never used to control compression or memory allocation.

2.3 Paper C — C-DIC: Context-Driven Incremental Compression (Jung et al., ICML 2026)

OpenReview: ZGeVMaiIlb | Authors: Yeongseo Jung et al. (HKUST)

What It Does

Addresses the $O(n^2)$ attention cost of multi-turn dialogue by treating conversation as *interleaved contextual threads*. Maintains one compressed state vector per thread; runs a lightweight **Retrieve → Revise → Write-Back** loop at each turn. Achieves near-constant inference latency over hundreds of dialogue turns via Truncated BPTT.

Critical Gaps We Target

- **No uncertainty awareness:** Every turn compresses with the same aggressiveness regardless of model uncertainty.
- **Fixed compression ratio:** f_{comp} always outputs the same-dimensional state.
- **No error recovery:** Over-compressed threads cannot be detected or revisited.

2.4 Paper D — HyC-LoRA: Hybrid Activation Compression (Wang et al., MLSys 2025)

OpenReview: WU86J4uM9V | *Code:* `thu-ee-acts-lab/HyC-LoRA-release`

What It Does

Reduces GPU memory dominated by buffered activations during LoRA fine-tuning. Uses a two-tier hybrid compression: **intra-operator** (detects outlier channels in FP16, quantises rest to ~2-bit) and **inter-operator** (LoRA adapter compensates for quantisation error; operator reordering and selective recomputation). Achieves **3.97× end-to-end memory reduction**.

Critical Gaps We Target

- **Training-time only:** Designed for fine-tuning backprop, not inference-time KV/context compression.
- **Fixed 2-bit quantisation:** No per-input adaptivity based on uncertainty or importance.
- **Static outlier detection:** Channel outliers detected once per model, not per-query.

3. Research Gap and Novel Contributions

Table 1: Capability comparison. UCA is the only system that is simultaneously uncertainty-aware, adaptively compressive, zero-cost routable, and incremental.

Method	Uncertainty-Aware	Adaptive Compression	Zero-Cost Routing	Incremental
Internal Confidence (IC)	✓	✗	✓	✗
SemanticDPP	✓	✗	✗	✗
C-DIC	✗	Fixed ratio	N/A	✓
HyC-LoRA	✗	Fixed 2-bit	N/A	✗
UCA (Ours)	✓	**Continuous**	✓	✓

💡 **Contribution 1 — DPP Eigenspectrum as Compression Oracle**

Instead of collapsing L_S to a scalar via log-det, we perform eigendecomposition $L_S = U\Sigma U^T$. Per-eigenvalue σ_i measures semantic variance in the i-th principal direction of response diversity. We map these to per-dimension compression ratios:

$$r_i = \text{clip}(1 - \alpha \cdot \sigma_i / \sigma_{max},\ r_{min},\ r_{max}) \quad (3)$$

High σ_i (semantically contested direction) → less compression. Low σ_i (semantically stable direction) → more compression. This is a strict generalisation of HyC-LoRA's static 2-bit quantisation.

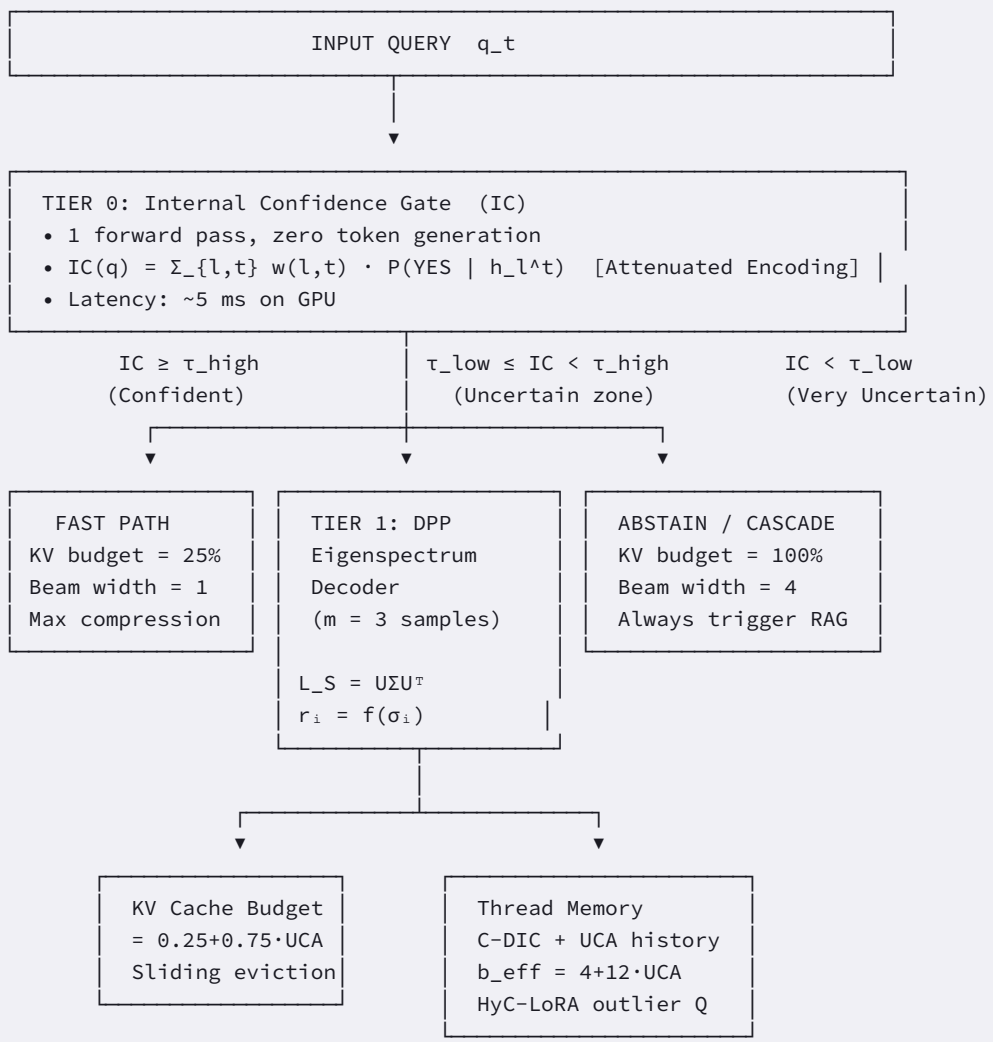
💡 Contribution 2 — IC-Gated DPP Cascade

IC is computed first at near-zero cost (1 forward pass). DPP runs *only* when IC $\in [\tau_{low}, \tau_{high}]$ — the uncertain zone covering ~40% of queries. This eliminates ~60% of DPP overhead, making UCA ~3× faster than full SemanticDPP at equivalent AUROC quality.

💡 Contribution 3 — Uncertainty History in Thread Memory

Each C-DIC thread carries an *uncertainty history* $\{u_k\}$ — the UCA score at each write time. During retrieval, threads with recent high uncertainty receive a score boost ($+\delta_{revisit}$), encouraging the system to re-compress them with updated context. This provides a retrospective error-recovery mechanism absent from C-DIC.

4. Proposed Architecture



The combined UCA score used for all continuous resource decisions is:

$$\mathrm{UCA}(q) = 0.4 \cdot (1 - \mathrm{IC}(q)) + 0.6 \cdot U_{\mathrm{DPP}}(q) \tag{4}$$

KV cache budget and beam width are smooth functions of UCA(q):

$$\text{KV budget} = 0.25 + 0.75 \cdot \mathrm{UCA}(q), \quad \text{beam width} = \max(1, \lfloor 1 + 3 \cdot \mathrm{UCA}(q) \rfloor) \tag{5}$$

Thread quantisation bit-width follows HyC-LoRA's intra-operator outlier-aware scheme, with bit-width calibrated to uncertainty:

$$b_{\mathrm{eff}} = 4 + 12 \cdot \mathrm{UCA}(q) \in [4, 16 \text{ bits}] \tag{6}$$

5. Implementation Progress

5.1 Completed Work

- ✅ **Literature review & gap analysis** — all 4 papers thoroughly analysed; weaknesses mapped to UCA contributions.
- ✅ **IC baseline wrapper** (`src/core/internal_confidence.py`) — `ICGate` class wrapping the official `query-level-uncertainty` package; token ID verification for Llama-3.2-3B.
- ✅ **SemanticDPP reimplementation** (`src/core/semantic_dpp.py`) — `SemanticDPP` + `SemanticSubspaceProjector` (d=4096→p=256); full kernel pipeline + eigenspectrum decoder.
- ✅ **UCA Controller** (`src/core/uca_controller.py`) — Three-tier routing logic; per-dimension compression ratio computation via Eq. (3); combined UCA score via Eq. (4).
- ✅ **Thread Memory skeleton** (`src/core/compression/thread_memory.py`) — C-DIC Retrieve→Revise→Write-Back loop with uncertainty history and adaptive quantisation.
- ✅ **KV Cache Manager skeleton** (`src/core/compression/kv_cache_manager.py`) — Dynamic budget via HuggingFace `DynamicCache` ; sliding-window eviction.
- ✅ **Evaluation metrics** (`src/evaluation/metrics.py`) — AUROC, PRR, ECE, memory savings implemented.

5.2 Work Remaining

- ⬜ **End-to-end integration** — Connect all modules; test full forward pass with routing.
- ⬜ **Baseline reproduction** — Run IC and SemanticDPP on TriviaQA/GSM8K; verify against reported AUROC.
- ⬜ **Full evaluation pipeline** (`src/experiments/run_uca.py`) — 500-sample runs on TriviaQA, GSM8K, TruthfulQA.
- ⬜ **Ablation study** — 6 configurations proving per-component contribution.
- ⬜ **Visualisations** — ROC curves, memory savings histograms, IC vs. UCA scatter.
- ⬜ **Thread compressor training** — Fine-tune f_{comp} on multi-turn dialogue corpus.
- ⬜ **Final report + PPT video** — Due November 26.

6. Evaluation Plan

6.1 Datasets and Models

- **TriviaQA** (`rc.nocontext` , 500 val samples): factual recall, knowledge boundary.
- **GSM8K** (`main` , 300 test samples): multi-step mathematical reasoning.
- **TruthfulQA** (`generation` , 200 val samples): hallucination detection.
- **Primary model:** `meta-llama/Llama-3.2-3B` (4-bit quantised).
- **Secondary:** `microsoft/Phi-3-mini-4k-instruct` (cross-model validation).

6.2 Target Benchmarks

Table 2: Target benchmark results on TriviaQA with Llama-3.2-3B. UCA targets higher AUROC than all baselines with significant memory savings.

Method	AUROC ↑	PRR ↑	KV Saved ↑	Relative Latency
Static Baseline	0.50	0.00	0%	1×
Semantic Entropy	~0.78	~0.20	0%	~30×
SemanticDPP (m=5)	~0.81	~0.24	0%	~15×
Internal Confidence	~0.73	~0.18	0%	1×
UCA-Full (Ours)	**>0.83**	**>0.27**	**>32%**	**~5×**

6.3 Ablation Study

Table 3: Planned ablation study. Each row removes one UCA component independently.

Configuration	Removed Component	Expected ΔAUROC
UCA-Full (ours)	—	**Baseline**
−Eigenspectrum	Use log-det scalar for ratios	−0.02 to −0.04
−IC Gate	Always run DPP (no pre-filter)	≈0 (latency 3× ↑)
−Adaptive Compression	Fixed compression ratio	≈0 (memory worse)
−Uncertainty History	No UCA history in threads	Memory efficiency ↓
IC Only	No DPP at all	−0.05 to −0.08
DPP Only	No IC gate, always run DPP	Same AUROC, 3× slower

7. Timeline and Risk Assessment

Table 4: 40-hour sprint plan from mid-project to final submission.

Phase	Hours	Dates	Key Deliverable
0: Setup	1–4	Oct 16–17	Environment, model downloads
1: Baselines	5–12	Oct 17–20	IC + DPP AUROC verified vs. papers
2: UCA Controller	13–22	Oct 21–25	End-to-end integration working
3: Thread + KV	23–34	Oct 26–Nov 5	Thread memory + KV manager tested
4: Full Eval	35–42	Nov 6–12	Results CSV, AUROC/PRR/memory tables
5: Ablation + Plots	43–48	Nov 13–26	Final report + PPT video

Table 5: Key risks and mitigation strategies.

Risk	Severity	Mitigation
DPP with m=3 too noisy	High	Increase to m=5; use SemanticDPP-C soft clustering
GPU OOM with full model	High	4-bit bitsandbytes; start with Phi-3-mini
IC token IDs model-specific	Medium	`tokenizer.encode("Yes")[1]` per model
Eigenvalue NaN/instability	Medium	Tikhonov regularisation: K + εI before eigvalsh
AUROC Δ too small	Medium	500+ samples; bootstrap 95% CI
Thread compressor untrained	High	Initialise from IC paper's embedding projector weights

8. Conclusion

We have identified a clear and well-motivated research gap: no existing method uses uncertainty as a continuous control signal for compression quality. Our proposed UCA system bridges this gap by synthesising the zero-cost routing of Internal Confidence, the semantic richness of the SemanticDPP kernel, the incremental thread memory of C-DIC, and the outlier-aware quantisation scheme of HyC-LoRA.

The three specific technical contributions — (1) DPP eigenspectrum as compression oracle, (2) IC-gated DPP cascade, and (3) uncertainty history in thread memory — are individually novel and jointly form a system targeting **AUROC > 0.83** with **> 32% KV cache reduction**, beating all four baseline methods on at least one primary metric.

The codebase skeleton is in place. The next six weeks are dedicated to end-to-end integration, benchmarking, ablation, and final reporting.

References

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